Chapter 3: Angular Momentum - Detailed Answer Key

Educator Note: This key assumes the standard convention where counter-clockwise rotation is positive (+) and clockwise rotation is negative (-), unless specified otherwise by the problem context. N denotes rotational speed in revolutions per second (r.p.s.), and ω (or θ') denotes angular velocity in radians per second (rad/s). Recall that $\omega = 2\pi N$.

Exercise 5

Concept: Angular Momentum of a Rigid Body ($\sigma = I\theta'$)

Given:

- Mass of hoop (m) = $200 \text{ g} = 0.2 \text{ kg}$
- Radius (r) = $20 \text{ cm} = 0.2 \text{ m}$
- Rotational Speed (N) = 10 r.p.s. in the **negative** direction.
- Moment of Inertia for hoop (I) = mr^2

Analysis:

1. Calculate Angular Velocity (θ'):

$$\theta' = -2\pi N = -2\pi(10) = -20\pi \approx -62.83 \text{ rad/s}$$

2. Calculate Moment of Inertia (I):

$$I = mr^2 = 0.2 \times (0.2)^2 = 0.008 \text{ kg} \cdot \text{m}^2$$

3. Calculate Angular Momentum (σ):

$$\sigma = I\theta'$$

Calculation:

$$\sigma = 0.008 \times (-62.83) = -0.5026 \text{ kg} \cdot \text{m}^2/\text{s}$$

Answer: $-\mathbf{0.503} \text{ kg} \cdot \text{m}^{\mathbf{2}}/\text{s}$ (or -0.16π)

Exercise 6

Concept: Angular Momentum of a System ($\sigma_{\text{sys}} = \sum \sigma_i$)

Given:

- Particle 1: $I_1 = 0.1 \text{ kg} \cdot \text{m}^2$, $\theta'_1 = 2 \text{ rad/s}$
- Particle 2: $I_2 = 0.3 \text{ kg} \cdot \text{m}^2$, $\theta'_2 = -4 \text{ rad/s}$ (opposite direction)

Analysis: The total angular momentum is the algebraic sum of individual momenta.

Calculation:

$$\sigma_{\text{sys}} = (I_1 \times \theta'_1) + (I_2 \times \theta'_2)$$

$$\sigma_{\text{sys}} = (0.1 \times 2) + (0.3 \times -4)$$

$$\sigma_{\text{sys}} = 0.2 - 1.2 = -1.0 \text{ kg} \cdot \text{m}^2/\text{s}$$

Answer: $-\mathbf{1.0} \text{ kg} \cdot \text{m}^{\mathbf{2}}/\text{s}$

Exercise 7

Concept: Conservation of Angular Momentum (Change in Internal Mass Distribution)

Given:

- Initial Speed (N_1) = 0.50 r.p.s.
- Initial Inertia (I_1) = $13.10 \text{ kg} \cdot \text{m}^2$
- Final Inertia (I_2) = $5.225 \text{ kg} \cdot \text{m}^2$

Analysis: Since external torque $\sum M_{\text{ext}} = 0$, angular momentum is conserved: $\sigma_1 = \sigma_2$.

$$I_1\theta_1' = I_2\theta_2' \implies I_1(2\pi N_1) = I_2(2\pi N_2) \implies I_1N_1 = I_2N_2$$

Calculation:

$$13.10 \times 0.50 = 5.225 \times N_2$$

$$6.55 = 5.225 \times N_2$$

$$N_2 = \frac{6.55}{5.225} \approx 1.253 \text{ r.p.s.}$$

Answer: $\mathbf{1.25}$ r.p.s.

Exercise 8

Concept: Conservation of Angular Momentum (Inelastic Rotational Collision)

Given:

- Disk: $m_d = 0.5 \text{ kg}$, $r_d = 0.4 \text{ m}$, $N_1 = 2 \text{ r.p.s.}$
- Particle: $m_p = 0.1 \text{ kg}$, added at distance $r_p = 0.2 \text{ m}$.
- Disk Inertia Formula: $I_{\text{disk}} = \frac{1}{2}m_d r_d^2$

Analysis:

1. **Initial State:** Only disk rotating.

$$I_{\text{disk}} = 0.5 \times 0.5 \times (0.4)^2 = 0.04 \text{ kg} \cdot \text{m}^2$$

$$\sigma_{\text{initial}} = I_{\text{disk}} \times 2\pi N_1 = 0.04 \times 4\pi = 0.16\pi$$

2. **Final State:** Disk + Particle rotating together.

$$I_{\text{particle}} = m_p r_p^2 = 0.1 \times (0.2)^2 = 0.004 \text{ kg} \cdot \text{m}^2$$

$$I_{\text{total}} = I_{\text{disk}} + I_{\text{particle}} = 0.04 + 0.004 = 0.044 \text{ kg} \cdot \text{m}^2$$

3. **Conservation:**

$$\sigma_{\text{initial}} = \sigma_{\text{final}}$$

$$0.16\pi = I_{\text{total}} \times 2\pi N_2$$

$$0.16\pi = 0.044 \times 2\pi N_2$$

Calculation:

$$0.08 = 0.044N_2$$

$$N_2 = \frac{0.08}{0.044} = \frac{80}{44} \approx 1.818 \text{ r.p.s.}$$

Answer: $\mathbf{1.82}$ r.p.s.

Exercise 9

Concept: Conservation of Angular Momentum (Coupling/Clutching)

Given:

- Flywheel 1 (R_1): $I_1 = 2.27 \text{ kg} \cdot \text{m}^2$, Speed = 2 r.p.s.
- Flywheel 2 (R_2): $I_2 = 3.52 \text{ kg} \cdot \text{m}^2$, Speed = 2 r.p.s.
- **Crucial:** They rotate in **opposite** directions.

Analysis: Let R_1 be positive and R_2 be negative.

$$\omega_1 = 2\pi(2) = 4\pi \text{ rad/s}$$

$$\omega_2 = -2\pi(2) = -4\pi \text{ rad/s}$$

1. **Initial Momentum:**

$$\sigma_{\text{tot}} = I_1\omega_1 + I_2\omega_2 = 2.27(4\pi) + 3.52(-4\pi)$$

$$\sigma_{\text{tot}} = 4\pi(2.27 - 3.52) = 4\pi(-1.25) = -5\pi \text{ kg} \cdot \text{m}^2/\text{s}$$

2. **Final Momentum:**

$$\sigma_{\text{final}} = (I_1 + I_2)\omega_f$$

$$-5\pi = (2.27 + 3.52)\omega_f$$

$$-5\pi = 5.79\omega_f$$

Calculation:

$$\omega_f = \frac{-5\pi}{5.79} \approx -2.712 \text{ rad/s}$$

$$N_f = \frac{|\omega_f|}{2\pi} = \frac{2.5}{5.79} \approx 0.431 \text{ r.p.s.}$$

Answer: $\mathbf{0.43}$ r.p.s.

Exercise 10

Concept: Conservation with Variable Radius

Given:

- Rod: $m' = 0.2$ kg, $L = 0.5$ m. $I_{\text{rod}} = \frac{m'L^2}{12}$.
- Block B: $m = 0.1$ kg.
- Initial State: $r_1 = 0.15$ m, $N_1 = 2$ r.p.s.
- Final State: $r_2 = 0.25$ m (End of rod, $L/2$).

Analysis:

1. **Inertia Calculations:**

$$I_{\text{rod}} = \frac{0.2 \times 0.5^2}{12} = \frac{0.05}{12} \approx 0.004167 \text{ kg} \cdot \text{m}^2$$

$$I_{\text{B1}} = mr_1^2 = 0.1 \times 0.15^2 = 0.00225 \text{ kg} \cdot \text{m}^2$$

$$I_{\text{B2}} = mr_2^2 = 0.1 \times 0.25^2 = 0.00625 \text{ kg} \cdot \text{m}^2$$

2. Conservation:

$$(I_{\text{rod}} + I_{\text{B1}})N_1 = (I_{\text{rod}} + I_{\text{B2}})N_2$$

$$(0.004167 + 0.00225) \times 2 = (0.004167 + 0.00625) \times N_2$$

$$0.006417 \times 2 = 0.010417 \times N_2$$

Calculation:

$$N_2 = \frac{0.012834}{0.010417} \approx 1.232 \text{ r.p.s.}$$

Answer: $\mathbf{1.23}$ r.p.s.

Exercise 12

Concept: Dynamics ($\sum M = I\theta''$) and Work-Energy

Given:

- Hoop: $m = 150$ kg, $r = 0.2$ m. $I = mr^2$.
- Force: $F = 45$ N (constant, tangential).
- Starts from rest ($t = 0, \omega_0 = 0$).

a) Angular Acceleration (θ'' or α)

- Torque $M = F \times r = 45 \times 0.2 = 9 \text{ N} \cdot \text{m}$.
- Inertia $I = 150 \times 0.2^2 = 6 \text{ kg} \cdot \text{m}^2$.
- $\theta'' = \frac{M}{I} = \frac{9}{6} = \mathbf{1.5} \text{ rad/s}^{\mathbf{2}}$.

b) Angular Velocity at $t = 4$s & Revolutions

- $\omega = \omega_0 + \theta'' t = 0 + 1.5(4) = \mathbf{6}$ rad/s.
- Angle $\theta = \frac{1}{2}\theta'' t^2 = 0.5(1.5)(4^2) = 12$ rad.
- Revolutions $= \frac{12}{2\pi} \approx \mathbf{1.91}$ revs.

c) Work Done vs. Kinetic Energy

- Work $W = M \cdot \theta = 9 \text{ N} \cdot \text{m} \times 12 \text{ rad} = \mathbf{108}$ J.
- Kinetic Energy $K = \frac{1}{2}I\omega^2 = 0.5 \times 6 \times (6)^2 = 3 \times 36 = \mathbf{108}$ J.
- **Verification:** $W = \Delta K$. The theorem holds.

Exercise 14

Concept: Central Forces and Angular Momentum

a) Force Exerted by Observer The force is provided by the tension in the string, which acts as the centripetal force maintaining the circular motion.

$$F_c = ma_n = mR(\theta_1')^2$$

Answer: $\mathbf{F = mR\theta_1'^{\,2}}$

b) New Angular Velocity at $r = R/2$ The force is central (passes through the axis), so the torque is zero. Angular momentum is conserved.

$$\sigma_1 = \sigma_2$$

$$mR^2\theta_1' = mr^2\theta_2'$$

$$mR^2\theta_1' = m(\frac{R}{2})^2\theta_2'$$

$$R^2\theta_1' = \frac{R^2}{4}\theta_2'$$

$$\theta_2' = 4\theta_1'$$

Answer: The new angular velocity is **4 times the initial angular velocity**.

Exercise 16

Concept: Interaction between System Components (Action-Reaction)

Given:

- Man (m) = $100\ \mathrm{kg}$.
- Disk Radius (R) = $2\ \mathrm{m}$.
- Disk Inertia (I_D) = $4000\ \mathrm{kg} \cdot \mathrm{m}^2$.
- System initially at rest.
- Man's velocity **with respect to Earth** (v_m) = $1\ \mathrm{m/s}$.

a) Disk Speed and Direction Conservation of Angular Momentum (System $\sigma_{\text{final}} = 0$).

$$\sigma_{\text{man}} + \sigma_{\text{disk}} = 0$$

$$(mv_mR) + (I_D\omega_D) = 0$$

$$(100 \times 1 \times 2) + 4000\omega_D = 0$$

$$200 + 4000\omega_D = 0$$

$$\omega_D = -\frac{200}{4000} = -0.05\ \mathrm{rad/s}$$

Answer: The disk rotates at **0.05 rad/s** in the **opposite direction** to the man's motion.

b) Angle of Rotation The man returns to his initial position **with respect to Earth**. This implies he completes one full circle relative to the ground (2π radians).

- Man's angular velocity: $\omega_m = \frac{v_m}{R} = \frac{1}{2} = 0.5\ \mathrm{rad/s}$.
- Time to complete circle: $t = \frac{2\pi}{\omega_m} = \frac{2\pi}{0.5} = 4\pi\ \mathrm{s}$.
- Angle disk rotates in this time: $\theta_D = \omega_D \times t = -0.05 \times 4\pi = -0.2\pi\ \mathrm{rad}$.

Answer: The disk rotates by 0.2π **radians** (or 36°).